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Measuring Topics Using Cross-Domain Supervised Learning: Methods and Application to New Zealand Parliament*

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Abstract

This paper studies and assesses a novel method for assigning topics to political texts: cross-domain supervised learning. A machine learning algorithm is trained to classify topics in a labeled source corpus and then applied to extrapolate topics in an unlabeled target corpus. An advantage of the method is that, unlike standard (unsupervised) topic models, the set of assigned topics are interpretable and scientifically meaningful by construction. We demonstrate the method in the case of labeled party manifestos (source corpus) and unlabeled parliamentary speeches (target corpus). Besides the standard cross-validated within-domain error metrics, we further validate the cross-domain performance by labeling a subset of target corpus documents. We find that the classifier assigns topics accurately in the parliamentary speeches, although accuracy varies substantially by topic. To assess the construct validity, we analyze the impact on parliamentary speech topics of New Zealand’s 1996 electoral reform, which replaced a first-past-the-post system with proportional representation.

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1 Introduction

Social scientists have expended significant resources to hand-code political text data. For example, the Comparative Agendas Project and the Comparative Manifesto Project have coded many documents across a variety of politically relevant categories (Budge et al., 2001; John et al., 2013; Jones and Baumgartner, 2005; Klingemann et al., 2006). Meanwhile, an increasing number of studies hand-code a subsample of text data and then use supervised learning to automatically code the full sample of unlabeled documents (Hopkins and King, 2010; Workman, 2015). Among the studies using hand-coded documents are studies on party competition, legislative politics and political stability (e.g., Tsebelis, 1999; Adams et al., 2006; Tavits and Letki, 2009; Böhmelt et al., 2016).

In this paper, we study and assess cross-domain supervised learning, a novel method to measure topics in political text data. This technique enables researchers to use existing hand-coded data from a specific domain, such as the Comparative Manifesto Corpus, to infer categories of text data from a different domain, such as parliamentary speeches. Cross-domain supervised learning has several advantages in comparison to existing methods for classifying topics in political science (Denny and Spirling, 2018; Wilkerson and Casas, 2017). In comparison to supervised learning, cross-domain supervised learning significantly reduces the costs of data collection because researchers can use existing data. A major advantage over dictionary methods and topic models is that we can use the standard test-sample metrics from machine learning to see how well the classification system works (Hastie, Tibshirani and Friedman, 2009). This is important for assessing the validity of downstream empirical results.

We assess cross-domain supervised learning using data on party platforms from the Comparative Manifesto Project and parliamentary speeches from New Zealand. These corpora serve similar communication goals: parties use platforms, and parliamentarians use speeches, to advance their policy ideas and their re-election prospects (e.g., Martin and Vanberg, 2008; Proksch and Slapin, 2014). However, the distribution of words in manifestos and speeches may also differ because the documents stem from a different communication context. Hence, it is important to assess and validate the model predictions.

Methodologically, we start with training a machine classifier based on the manifesto corpus. The corpus includes over 115,000 manifesto statements labeled according to 8 broad topics and 44 narrow topics. In classifying the 44 narrow topics, the model achieves out-of-sample classification accuracy of 54 percent, more than double the baseline accuracy of picking the most frequent topic. In the aggregated 8-class model, the accuracy improves to 64 percent and we document good precision and recall across all classes.

With the trained topic predictor in hand, we use it to classify topics in a corpus of parliamentary speech transcripts from the New Zealand Parliament. This new corpus encompasses the universe of parliamentary speeches for the period 1987-2002 (nearly 300,000 speeches). To validate that the topic prediction works in the new domain, we compare predictions to those made by an expert coder for 4,165 parliamentary speeches. We find that the accuracy is similar to the expected accuracy inherent in human coder misclassification. We assess the replicability of our findings by asking three additional coders to code a subset of the speeches. For additional robustness, we show that the topic predictions have similar accuracy in speeches by U.S. Congressmen.

To assess the construct validity of the topics, we study the consequences of New Zealand’s 1993 electoral reform, which changed the system from first-past-the-post to mixed-member proportional representation. In contrast to first-past-the-post systems, mixed-member proportional representation systems facilitate the formation of coalition and minority governments, which involve principal-agent problems among coalition parties and tend to be less stable (e.g., [King et al., 1990](#); [Martin and Vanberg, 2005](#)). In line with the previous qualitative and quantitative evidence (e.g., [Duverger, 1957](#); [Powell, 2000](#); [Taagepera and Shugart, 1989](#); [Vowles et al., 2002](#); [Barker et al., 2003](#)), we find that the reform significantly increased attention toward political authority, which includes discussions about political (in)stability and party (in)competence.

This work fits into a growing methodological literature using text data to analyze dimensions of political discourse, reviewed in Section 2. Section 3 presents the cross-domain supervised learning approach. In Section 4, we present the classification results and validation in the target corpus. Section 5 applies the technique to an analysis of the electoral reform in New Zealand. Section 6 concludes.

2 Background: Topic Classification in Political Science

Text data is difficult to analyze with traditional empirical methods due to its lack of structure and high dimensionality. To analyze political topics or policy issues in text, for example, one must first assign documents to categories. There are three main approaches to this problem: lexicon-based pattern matching, unsupervised topic models, and supervised learning classifiers. This section discusses the pros and cons of these methods, as well as how our new method (cross-domain supervised learning) fits in.

Table 1 provides a summary of the different approaches and the key design factors that researchers should take into account when deciding on the approach. [Quinn et al. \(2010\)](#)

Table 1: Summary of Design Factors for Topic Classification Methods

	Dictionaries (Custom)	Dictionaries (Generic)	Topic Modeling	Within-Domain Supervised Learning	Cross-Domain Supervised Learning
Design Efficiency	Low	High	High	Low	High
Annotation Efficiency	High	High	High	Low	Moderate
Specificity	High	Moderate	Low	High	Moderate
Interpretability	High	High	Moderate	High	High
Validatability	Low	Low	Low	High	High

provide a similar summary of assumptions and costs in their introduction of unsupervised topic models to political science. We build on that perspective to highlight the appropriate use of cross-domain learning.

The classification methods are evaluated along five factors. First, design efficiency assesses the expert researcher time needed in designing the classification system. Second, annotation costs summarize the non-expert time needed, notably for annotating documents. Third, specificity refers to how much the system can be targeted toward answering particular questions or exploring particular features in the data. Interpretability summarizes how straightforward it is to interpret the resulting topic classifications. Finally, validatability refers to the feasibility of validating topics – that is, to check whether the classifier is producing topics correctly.

The dictionary or lexicon-based approach works by searching for particular textual patterns in the text to assign topics. Researchers can use either existing, generic dictionaries (e.g., [Crabtree et al., 2019](#)), such as LIWC ([Pennebaker et al., 2015](#)), or create their own custom dictionaries. For example, one could look for the word “taxes” (and its variants) to detect documents about taxes, for the phrase “women” to detect documents about gender issues, and so on. [Laver, Benoit and Garry \(2003\)](#) explore a word-based approach to analyze ideological positions and provide uncertainty measures using an annotated sample.

Custom dictionaries have significant up-front costs for the researcher to build the tags dictionary, but after that the annotation costs are zero. They have high specificity in the sense that they give the researcher full control over the dimensions of text they would like to target. If one is interested in taxes, one can search for that topic. The method is also highly

interpretable because the tags already contain expert knowledge and can be inspected easily.

Like custom dictionaries, generic dictionaries have the advantages of negligible annotation costs and high interpretability. One can easily read the full list of terms to see what is going on. The advantage of generic dictionaries is the much lower up-front design time, as they are already produced and validated by previous researchers. The tradeoff is a significant loss in specificity, as one can only measure the dimensions of text that are available in generic dictionaries.

For both custom and generic dictionaries, the major downsides are in their highly constrained representation of language and limited validatability. The lexicon tags are unavoidably subjective, over-inclusive, and under-inclusive. Politicians might use the word “revenues” to refer to taxes, for example, and they might use the word “women” in many contexts unrelated to gender policies. Some documents will have tags from multiple categories, and many documents will have no tags. There are no easy ways to deal with these cases. One cannot tell how well the labels work without significant investment in labeling of documents, which defeats the major advantage of dictionaries (the low annotation costs).

The next major approach to text classification is topic modeling, such as latent dirichlet allocation (LDA) (Blei, Ng and Jordan, 2003), non-negative matrix factorization (Lee and Seung, 1999), and structural topic model (Roberts et al., 2013). These algorithms provide a form of interpretable dimension reduction, where documents are transformed from high-dimensional counts over words to low-dimensional shares over topics. Topic models are a powerful tool because they often produce intuitive, interpretable topics without any labeled training data (Lucas et al., 2015; Greene and Cross, 2017). Example applications of topic models in political science include Quinn et al. (2010) and Grimmer (2010), who use them to analyze political communications by U.S. Congressmen.¹

The major advantage of topic models, as mentioned, is that they do not require any labeled training data to classify documents into categories. The design costs are very low, as for example in LDA the only major design choice is how many topics, and most other steps are automated. In turn, there are zero annotation costs as a trained topic model can instantly produce a set of topic shares for any given document. It allows documents to have multiple topics, and properly deals with all documents.

However, topic models have significant limitations. The topics are learned directly from the data, so specificity is very low and the topics may or may not represent the language dimensions in which the researcher is interested. While the topics produced can be useful

¹For the purposes of this methodological overview, document embedding approaches such as paragraph vectors (Quoc and Mikolov, 2014) can also be considered as topic models. For an example application, see Demszky et al. (2019).

and interpretable, that often requires significant tweaking.

These issues are amplified by the problem that the produced topics can be quite sensitive to perturbations in the data, for example the steps taken in text preprocessing and featurization (Denny and Spirling, 2018). Changing the number of topics often can completely change the character of the topics. Even changing the order of documents can have a big impact. Finally, there is a subjectivity issue as researchers have to interpret the content of the topics *ex post* (Wilkerson and Casas, 2017).

These issues of tweaking and the subjectivity of interpretations invite specification search. This problem is significant because there are no unified validation standards for topic models (Lowe and Benoit, 2013).² Validatability is a major challenge for topic models because that would require annotation of documents, which precludes the efficiency benefits from unsupervised learning.

A third approach, supervised learning of topics, can address the major shortcomings of dictionary methods and topic models. The defining feature of this approach is that researchers randomly sample some of the documents and hand-annotate the topics to create a labeled training dataset (e.g., Hopkins and King, 2010; Drutman and Hopkins, 2013; Workman, 2015). With a set of labeled documents in hand, one can use machine learning to encode the relations between text features and topics. The trained machine learning model can then automatically classify the topics in unlabeled data. For example, Benoit et al. (2019) use hand-coded manifesto data to train a supervised learning model and then recode manifesto data to new categories.³

Supervised learning has several major advantages. First, like custom dictionaries the system can be highly targeted toward classifying any dimension of the text that the researcher is interested in (high specificity). Second, the workings of the system are highly interpretable because one can read the codebook provided to annotators. One can also look at example documents for each category. Third, and perhaps most importantly, supervised learning classifiers can be rigorously validated. In the standard machine learning approach, one divides up the annotated data into a training set and test set and assesses how well the classifier works in held-out test data (e.g. Hastie, Tibshirani and Friedman, 2009). The classification accuracy metrics in the test set provide a good estimate for how well the classifier will work in the unlabeled documents.

²See Herzog, John and Mikhaylov (2018) for a recent proposal for interpreting the topics of unsupervised learning models.

³For our purposes, text similarity measures can also be seen as a form of supervised learning. The literature has measured other dimensions of political text besides policy topics, such as polarization or policy diffusion (Wilkerson, Smith and Stramp, 2015; Peterson and Spirling, 2018; Goet, 2019; Greene, Park and Colaresi, 2019; Yan et al., 2019).

But again we face tradeoffs. The significant problem with supervised learning approach is their costs in terms of both design and annotation. First, the researcher has to design a set of topics and build detailed documentation and codebooks for annotators. Second, the annotators must spend significant time being trained, doing the actual annotations, and comparing results with other annotators. For most applications, it requires a large investment of time and money to classify enough documents to make a classifier useful.

That brings us to the task of this paper: assessing the use of cross-domain supervised learning. That is, we would like to use a text classifier built in one domain (the source corpus) and then apply it in another domain (the target corpus). Analogous to the move from custom dictionaries to generic dictionaries, the cross-domain approach inherits some of the benefits of within-domain supervised learning. In terms of interpretability and validatability, cross-domain learning is equivalent in its high marks to within-domain supervised learning. The classifier is interpretable in the same way, as one can read the annotation codebooks and examine example documents (but this time in the target corpus). The classifier is also validatable in the same way, especially if one annotates a sample of documents in the target corpus for computing cross-domain test-set accuracy metrics.

The main advantage of cross-domain learning over within-domain learning is that researchers can draw on existing labeled corpora as training data. This reduces the design costs to zero, as one borrows in full the schema and codebooks of the original system. Similarly, the annotation costs are largely eliminated as the previous annotations are used to train the classifier.

That said, some annotations are needed in cross-domain learning (as reflected in Table 1), not to build the classifier but to validate it. A sample of annotated documents in the target corpus are used to produce classification accuracy metrics, to make sure that the system is working as intended. The other major tradeoff, again analogous to the shift from custom to generic dictionaries, is a loss of specificity relative to within-domain supervised learning. The set of questions and policies that one can analyze with cross-domain supervised learning is relatively circumscribed, because one has to use existing labeled datasets.

Two recent papers have increased our understanding of cross-domain supervised learning in political science. The closest paper is [Burscher, Vliegthart and De Vreese \(2015\)](#), who hand-code 11,089 Dutch-language news articles and 4,759 parliamentary questions from the Dutch parliament to 19 issue categories from the Comparative Agendas Project. The paper explores cross-domain performance and relevance of different text featurization techniques. Second, [Yan et al. \(2019\)](#) train a classifier to predict whether U.S. Congress speeches are given by a Republican or Democrat, then apply the classifier to predict whether an article was written in a liberal or conservative media platform. The authors find that cross-domain

supervised learning does not work well because the concept of partisanship differs across these domains.

Our contribution lies in demonstrating and assessing the opportunities of cross-domain supervised learning in a setting that is very relevant for applied political scientists. More specifically, our study is, to our knowledge, the first paper that systematically implements cross-domain supervised learning of policy topics on the basis of the manifesto corpus, one of the largest hand-annotated corpora in political science. We use the manifesto corpus to predict topics in parliamentary speeches of the New Zealand Parliament, and demonstrate how those predictions can be used for a substantive empirical analysis.

The application of cross-domain-supervised learning to manifesto and speech data is especially promising for three reasons. First, our training and target corpora are both political texts and both are prepared by party actors. Manifesto and speech texts are strongly related because parliamentarians aim to implement the manifesto commitments in their parliamentary work. Second, the manifesto corpus is coded at the quasi-sentence level rather than the document-level. While entire speeches include multiple topics and rhetoric elements, our training data is fine-grained. Third, we can systematically assess the validity of cross-domain supervised learning by asking coders from the manifesto projects to hand-code speeches to manifesto categories. Hence, we make sure that we assess the machine predictions in line how the topics are conceptualized by the manifesto project.

3 Methods: Cross-Domain Supervised Learning

This section outlines our method for cross-domain supervised learning. In outline, we train a topic classifier in an annotated source corpus and apply it to an unlabeled target corpus, with a set of validation steps to check that meaningful topics are recovered in the target corpus. In our application, we train a machine classifier on the Comparative Manifesto Corpus to predict topic codes from manifesto statement text. Then we show how to use the trained classifier to predict policy topics across domains in a corpus of New Zealand parliamentary speeches.

3.1 Source Corpus: Comparative Manifesto Project

Our source corpus is a corpus of party platforms annotated by the Comparative Manifesto Project. We access the corpus via the Manifesto R package (Budge et al., 2001; Klingemann et al., 2006; Lehmann et al., 2017). We focus on English manifesto statements from Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States. This data set

has $N_S = 115,410$ rows of annotated policy statements, where S indicates the “source” corpus and documents are indexed by i .

Each statement includes a hand-annotated topic code. The statement “and reduce global warming emissions” refers, for example, to the environment (category 501), while the statement “We can’t afford another dose of Labour” to political authority (category 305). The manifesto project usually has one trained coder for each country.

We pre-process the topic codes k following two specifications. For the 44-topic specification ($K = 44$), we take into account all topics and merge categories that focus on the same topic but a different direction (positive/negative). For example, we put together categories “per607 Multiculturalism: Positive” and “per608 Multiculturalism: Negative” to create one “Multiculturalism” topic. This procedure leads to a sample of 44 categories. For the 8-topic specification ($K = 8$), we merge all categories to 8 major topics⁴ following the manifesto codebook (Budge et al., 2001; Laver and Budge, 1992). We have the following 8 major topics: external relations, freedom and democracy, political system, economy, welfare and quality of life, fabric of society, social groups, and no topic.

The topic label for document i is a K -vector $\mathbf{y}_i$ of indicator variables with items y_i^k for each topic k , equaling one if a document is annotated as topic k and zero otherwise.

3.2 Machine Classification of Policy Topics

The next step is to train a machine learning model to predict the manifesto topic codes. First we transform the plain text of the statements to a bag-of-n-grams representation. The model then takes the N-gram frequencies as feature inputs and outputs a probability distribution over topics.

First we featurize the text by implementing several standard text-preprocessing steps. We remove stopwords, punctuation, and capitalization to regularize the corpus. Using the processed tokens, we constructed N-grams up to length three: words, bigrams, and trigrams.

To build the vocabulary, we counted the term occurrences for each N-gram. We dropped N-grams appearing in fewer than 10 documents, as they would not contain much predictive information. We also dropped N-grams appearing in more than 40 percent of documents, as these are likely specific to manifestos and not distinctive of specific topics.

Finally, we computed term-frequency/inverse-document-frequency (TF-IDF) weights for each N-gram, where each statement is treated as a document.⁵ The resulting feature set has

⁴Called “domains” in the manifesto codebook.

⁵While N-grams model are known to work well for supervised learning, we got similar within-domain performance from taking the average of word embedding vectors in each document (Arora, Lian and Ma,

$M = 19,734$ columns, with each column indexed by j .

A range of machine classification models could be used. We got the best performance with a regularized multinomial logistic regression model, widely used in un-ordered multiclass prediction problems (Hastie, Tibshirani and Friedman, 2009; Géron, 2017). Formally, the model represents the probability that document i is on topic k as

$$\hat{p}_i^k = \frac{\exp(\theta_k \cdot \mathbf{x}_i)}{\sum_{l=1}^K \exp(\theta_l \cdot \mathbf{x}_i)}, \quad (1)$$

where $\mathbf{x}_i$ is the M -vector of n -gram frequencies for document i and θ_k is an M -vector of learnable parameter weights across features for topic $k \in \{1, \dots, K\}$.

Let Θ give the $M \times K$ matrix of parameters θ_{jk} for each feature j and each topic k . The classifier learns $\hat{\Theta}$ by minimizing the (regularized) categorical cross-entropy

$$\hat{\Theta} = \min_{\Theta} -\frac{1}{N_S} \sum_{i=1}^{N_S} \sum_{k=1}^K y_i^k \log(\hat{p}_i^k(\Theta)) + \lambda \sum_{j=1}^M \sum_{k=1}^K \theta_{jk}^2. \quad (2)$$

where the second term is the ridge (L2) penalty. The hyperparameter λ calibrates the strength of regularization and addresses overfitting.

We implemented the penalized logistic model using Python’s Scikit-Learn (Pedregosa et al., 2011). We used the newton-conjugate gradient solver and learned hyperparameters by three-fold cross-validation grid search in the training set. We trained separate models for the 8-topic and 44-topic labels. According to this procedure, the best parameters are an inverse of the regularization strength equal to two and no weighting of the categories (for both models).

As mentioned, a range of machine learning classifiers could be used to solve our problem. We experimented with other models, including a random forest, gradient boosting, and a neural net. The within-domain performance was similar for these models, but they have more hyperparameters to tune and take much longer to train. Thus, we decided to use the regularized multinomial logistic regression model for the rest of the analysis. In other contexts, we expect that using other text featurization and text classification methods would be preferred.

2016) and from producing pre-trained BERT embeddings (Devlin et al., 2018).

3.3 Target Corpus: New Zealand Parliament Speeches

Our target corpus consists of speeches given by members of the New Zealand Parliament for the period from 1987 until 2002. We extracted the speech data from the *Hansard*, which is the official record of the New Zealand Parliament.⁶ The data came in a series of zip archives, which included sets of files in HTML format.

We built the data as follows. First, a set of Python scripts parsed the HTML and extracted the speeches along with corresponding meta-data, most importantly the speaker. We identified 437,865 speeches in total and applied a set of filters as follows. We removed speeches given by the Speaker of the New Zealand Parliaments and his/her deputy because these parliamentarians are supposed to act in line with the general interest and not defend ideological or policy positions. Next, we removed short oral contributions held by ‘government member(s)’ and ‘opposition member(s)’ without further information on the name of the speakers. We also dropped speeches with fewer than 40 characters (excluding numbers) and Maori-language speeches for which an official translation was not provided.⁷ The final dataset has $N_T = 290,456$ documents, where T indicates the target corpus. This comprises 154,438 speeches and 136,018 questions.

We have detailed metadata from the *Hansard*, the New Zealand Parliamentary Information Service, and [Woldendorp, Keman and Budge \(2000\)](#). We identify the party membership, ministerial position, the election mechanism (list, constituency, Maori constituency) and the gender and ethnicity of the speaker. Furthermore, we collect data on whether the parliamentarian held a committee chairmanship. Regarding the type of speech, we know the date of the speech, the type of oral contribution (speech vs. question), and the stage of the speech (e.g., general debate, second reading debate).

3.4 Cross-Domain Prediction in Target Corpus

Finally, the text featurizer and text classifier trained in Subsection 3.2 can be applied to any snippet of text. Given an input document, it transforms the text to N-grams and outputs a set of predicted probabilities across topic classes. Here we explain how to use the text classification model, trained on the manifesto corpus, to predict topics in the New Zealand Parliament speech data.

Once our model is trained, applying it to a target corpus is straightforward. The TF-IDF vectorizer transforms each speech i into TF-IDF weighted N-gram frequencies $\mathbf{x}_i$, using the vocabulary and IDF weights from the source corpus. The logistic regression coefficients

⁶The data were distributed by the company *Knowledge Basket*.

⁷We identified and removed 31 speeches held in Maori language using the *polyglot* Python module.

produce a vector of predicted probabilities

$$\hat{p}_i^k = \frac{\exp(\hat{\theta}_k \cdot \mathbf{x}_i)}{\sum_{l=1}^K \exp(\hat{\theta}_l \cdot \mathbf{x}_i)}, \quad (3)$$

for each topic $k \in \{1, \dots, K\}$. The parameters $\hat{\Theta}$ are learned in the source corpus model.

We assign a single topic based on the highest-probability class. We do this for both the 44-topic and 8-topic specification. These predictions produce topics across all classes. For example, in the 8-topic model we obtain the following distribution in the New Zealand Parliament Speeches: economy (47,962 speeches), external relations (6,624), fabric of society (32,007), freedom and democracy (31,674), political system (94,766), social groups (18,262), welfare and quality of life (58,765) as well as no topic (396).

3.5 Target Corpus Annotations

As discussed, the advantage of cross-domain learning is that we use annotations in the source corpus to train a model and make predictions in the target corpus. Therefore target corpus labels are not needed for model training. However, they are needed for validation.

To that end, we arranged for the hand-coding of a sample of documents in the target corpus. We follow previous work that has used human judgment to validate automated classification of political documents (Lowe and Benoit, 2013). We hired the manifesto coder for New Zealand, who was trained by the manifesto project and has many years of experience in coding manifestos and other documents to manifesto categories.

The coder annotated a random sample of 4,165 parliamentary speeches to manifesto categories.⁸ The annotations took in total 52.5 hours.

To assess inter-coder reliability within the New Zealand target corpus (Mikhaylov, Laver and Benoit, 2012), we hired three additional coders.⁹ Like the main coder, these coders also received training from the manifesto project in English-language platforms. They were not experts on New Zealand politics, however. We drew a random sample of 250 speeches from the 4,165 speeches annotated by the first coder. Each of the three secondary coders annotated these same speeches, so that we had four annotations in total.

Finally, to assess broader generalization of the method, we also hand-annotated a corpus

⁸Note that we annotated topics by speech, not by sentence or by quasi-statement (as done in the standard comparative manifesto project approach). This was a design decision as our downstream empirical analysis is at the speech level, and it allowed us to obtain much more data than sentence-level annotations. The annotator had no trouble doing it this way. We only gave the coder the text of the speech and no meta-data such as the date or speaker.

⁹We thank Pola Lehmann for providing us the contact details of the manifesto coders.

of congressional speeches from the United States. We hired the manifesto coder for the United States and asked him to code a random sample of 150 speeches from the House of Representatives. The sample was drawn from all speeches contained in the *Congressional Record* for the period August 1987 through July 2002.¹⁰

3.6 Evaluating Model Performance

We follow the standard approaches in machine learning to evaluate the performance of our machine classifier. For the within-domain performance, we assess the predictive performance in a 25-percent held-out test sample. For the cross-domain performance, we compare the machine predictions to the new annotations provided by the human coder.

We report a variety of metrics to evaluate and understand model performance. First, we report the simple (Top 1) accuracy. This is the proportion of predicted topics (that is, the topic with the highest predicted probability) in the test set that are also the “true” topic as selected by human annotation. Note that simple accuracy is equal to the model’s micro-weighted aggregate precision, recall, and F1 score.

Especially for tasks with many classes, we might want to know not just how often the true topic is correctly ranked first, but also more broadly how often it is highly ranked and within the top few topics by predicted probability. In the Manifesto Corpus 44-topic specification, for example, there are some overlapping categories (e.g. economic goals and economic growth) which would likely be confused by either human or machine classifiers. To allow for this, we report the Top 3 (and Top 5) Accuracy. These give the proportion of observations for which the true class (from the hand annotations) is within the top three (or respectively top five) categories as ranked by their predicted probability from the machine classifier.

Another issue with simple accuracy is that they are summed across test samples, such that categories with more documents in the test sample are proportionally higher-weighted in the metrics. Therefore inaccurate predictions in the less frequent categories could be missed. To provide a more rounded aggregate report, we compute the balanced accuracy, which is the (un-weighted) average recall (fraction of true-class documents correctly identified) across output categories. Finally, we report the macro-weighted F1 score, which is the (un-weighted) average of the F1 Scores (harmonic mean of precision and recall) across all categories.

¹⁰In contrast to New Zealand, most short speeches in the U.S. House of Representatives are procedural. These speeches are used, for example, to yield speaking time to parliamentarians or report voting results. Similar to other parliamentary democracies such as United Kingdom, many short speeches in New Zealand take place in question times, which do not exist in the United States. We follow [Bäck, Debus and Fernandes \(2020\)](#) and limit our analysis of the United States to all speeches that have at least 50 words.

For any given empirical application, there are likely particular policy topics that the researcher is most interested in. Therefore we will also look at performance separately by class. Top 1 Accuracy gives the proportion of observations that are hand-annotated as a topic for which the machine places the highest probability on that topic. It is equivalent to the recall for a class. Top 3 Accuracy gives the percent of documents hand-coded to a category as being ranked by the classifier in the top three by predicted probability. Top 5 Accuracy gives the corresponding metric for the top five classes by predicted probability.

4 Classification Results

This section reports results on the performance of our classifier. We show that it works in-domain, in that it can reliably reproduce the hand-coded topic labels in the Manifesto Corpus. We also apply it to the New Zealand Parliament speeches and assess its performance in that second target domain.

4.1 Aggregate Performance

The classification results are summarized in Table 2. Columns 1 and 2 report results for the within-domain (source-source) predictions (manifesto-trained model applied to manifesto test corpus), while Columns 3 and 4 report results for the cross-domain (source-target) predictions (manifesto-trained model applied to the corpus of newly annotated New Zealand parliamentary speeches). Within each test corpus, we report metrics for 44 narrow topics (Columns 1 and 3) and 8 broad topics (Columns 2 and 4).

Starting with the 44-topic within-domain specification (Column 1), we see that the trained model predicts the correct category label 53.8 percent of the time. As expected, this

Table 2: Overview of classifier performance in test set

	(1)	(2)	(3)	(4)
	<u>Within-Domain</u>		<u>Cross-Domain</u>	
	44 topics	8 topics	44 topics	8 topics
Top 1 accuracy / F1 micro	0.538	0.641	0.410	0.507
Top 3 accuracy	0.763	0.908	0.649	0.816
Top 5 accuracy	0.839	0.971	0.746	0.916
Balanced accuracy	0.381	0.502	0.265	0.451
F1 macro	0.406	0.522	0.261	0.450

is worse than the training-sample prediction (71 percent accurate), as the model somewhat over-fits the training data. As there are 44 topic labels to be assigned, choosing randomly would be correct about 2 percent of the time. Choosing the top category would be correct 12 percent of the time. Meanwhile, there is significant human coder error in the training data (Mikhaylov, Laver and Benoit, 2012). Therefore 53.8 percent accuracy is quite good performance on the out-of-sample machine prediction task.

More numbers that demonstrate the within-domain efficacy of the classifier are seen in the Top 3 Accuracy (76.3 percent) and the Top 5 Accuracy (83.9 percent). These metrics show that even when the true class is not picked as having the highest probability, it is usually highly ranked. So if one is using predicted probabilities $\hat{p}_i^k$ in an empirical analysis, one can have some confidence that they contain information about textual variation in policy dimensions.

To qualify these statements, we also report balanced accuracy (0.381) and macro-weighted F1 (0.406). These worse numbers reflect that, perhaps unsurprisingly, the less frequent categories are more likely to be mis-classified. Empirical analyses of less frequent topics should be undertaken with caution.

In the 8-topic within-domain prediction (Column 2), we obtain better performance due to the smaller number of classes that the machine needs to assign. The test-sample accuracy is 64.1 percent is closer to the in-sample accuracy of 76 percent and significantly better than guessing randomly (12.5 percent accuracy) or guessing the top category (31 percent accuracy). The Top 3 Accuracy of 91 percent is similarly encouraging. As before, the balanced accuracy (0.502) and F1 score (0.52) indicate lower performance for the less frequent classes.

Examining the results on the 44 topics, we find an overall Top 1 accuracy of 0.41. This is significantly better than guessing at random (an accuracy of 2 percent) or guessing the most common hand-annotated class (accuracy of 19 percent). Perhaps unsurprisingly, the accuracy is lower than the in-domain accuracy (54 percent). We see even more encouraging performance when looking at top-3 accuracy (0.65) or top-5 accuracy (0.75).

At the bottom of the table, the metrics for the 8-topic specification are also relatively encouraging. The overall Top 1 performance is 51 percent, which is not that much less than the within-domain accuracy of 0.64. It is much better than guessing randomly (0.12) or the most common class (0.26). The top-3 and top-5 accuracies are 0.81 and 0.91.

In the aggregate, the cross-domain classifier seems to work. We can assign New Zealand speeches to policy topics based on the classifier trained on the Comparative Manifesto Corpus.

4.2 Performance by Topic

For most empirical applications, one would be interested in analyzing variation in particular topics. Therefore it is important to assess the variation in predictive performance across topics. To illustrate this type of evaluation, we report topic-level metrics. With relatively few broad topics, a confusion matrix works best. In the case of numerous topics, one can produce a table of accuracy metrics by topic. We illustrate both methods here.

Confusion matrices provides an intuitive visual report of model performance in specifications with relatively few classes (such that they can fit on a page easily). Here, we build confusion matrices in the 8-topic specification for the within-domain (Panel A) and cross-domain (Panel B) predictions.¹¹ In Table 3, rows index true categories while columns index predicted categories. In Panel A, a document is a test-set manifesto statement; in Panel B, a document is a hand-annotated New Zealand parliament speech. The number in a cell captures how often the model classified a document from the row class to the column class. The color (ranging from white to yellow to green) reflects relative frequency within-row, with a darker color meaning that this cell has more weight than other cells in the row.

As an illustrative example, the first row of Panel A shows that for the topic Economy, the within-domain model classified 5,270 of 7,197 manifesto statements correctly. 1,108 Economy statements were incorrectly classified as Welfare and Quality of Life, while 819 were incorrectly assigned to one of the other five categories (besides no topic). These numbers correspond to a topic-specific recall (top 1 accuracy) of 0.732, reported in the rightmost column.

Correspondingly, the Economy column reports the counts for each true topic that were (mis-)classified as Economy. For example, 1,033 Welfare documents and 608 Political System documents are mis-classified as Economy. This is perhaps not too surprising given the potential semantic overlaps in discussions of these topics. At the bottom of the column (and that of each topic), we report the ratio of total predicted count to total true count, which tells us how well the model replicates the distribution of topics. A value of 1.0 would mean that the distribution is the same; less than 1.0 means that the predictions are under-representative for this topic; greater than 1.0 means the predictions are over-representative for this topic. In the case of Economy, a predicted-to-true ratio of 1.11 means the predicted frequency of this topic is reasonably similar to the true frequency in the held-out test set.

Looking at Panel A as a whole, we can say that the within-domain model replicates well the annotated classes (besides the infrequent “Other Topic” class). The true category is selected most often across all topics. The minimum recall is a decent 0.42 (Social Groups), go-

¹¹Confusion matrices for the 44-topic task are included in Appendix Section 3.

Table 3: Classifier Performance with 8 Topics

A. Within-Domain Predictions for Party Platforms

	Economy	External Relations	Fabric of society	Freedom & Democracy	Political system	Social groups	Welfare & quality of life	No topic / Other	Total true	Recall
Economy	5270	93	131	40	301	254	1108	0	7197	0.732
External Relations	175	1207	137	83	85	49	209	1	1946	0.620
Fabric of society	269	107	1785	90	204	115	618	1	3189	0.560
Freedom and Democracy	102	60	135	631	219	35	177	0	1359	0.464
Political system	608	71	186	137	1255	65	542	1	2865	0.438
Social groups	493	51	185	29	111	1230	818	0	2917	0.422
Welfare and quality of life	1033	66	316	58	267	293	7138	0	9171	0.778
No topic / Other	58	6	37	9	34	7	55	3	209	0.014

Total predicted	8008	1661	2912	1077	2476	2048	10665	6
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Total predicted / Total true	1.11	0.85	0.91	0.79	0.86	0.70	1.16	0.03
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B. Cross-Domain Predictions for Parliamentary Speeches

	Economy	External Relations	Fabric of society	Freedom & Democracy	Political system	Social groups	Welfare & quality of life	No topic / Other	Total true	Recall
Economy	389	7	18	26	86	41	56	33	656	0.593
External Relations	8	53	4	4	14	3	0	5	91	0.582
Fabric of society	22	9	239	48	92	23	28	20	481	0.497
Freedom and Democracy	24	7	44	202	136	19	26	16	474	0.426
Political system	180	9	76	201	612	62	153	58	1351	0.453
Social groups	33	1	23	13	15	123	34	10	252	0.488
Welfare and quality of life	63	8	28	51	113	54	492	49	858	0.573
No topic / Other	1	0	0	0	0	0	0	1	2	0.500

Total predicted	720	94	432	545	1068	325	789	192
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Total predicted / Total true	1.10	1.03	0.90	1.15	0.79	1.29	0.92	96.00
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ing all the way up to 0.78 (Welfare and Quality of Life). The most common mis-classifications are somewhat intuitive. For example, many statements get put into the Welfare category, which could reflect that it is the most numerous category and is somewhat broad in its definition. Looking at the bottom row, meanwhile, we can see that overall the distribution of topics is replicated quite well. Economy and welfare are slightly over-represented, while the other categories (especially social groups) are somewhat under-represented.

Next we consider the cross-domain 8-topic confusion matrix in Panel B. The format of this matrix is the same as in Panel A, except that the predictions are made in the target corpus (New Zealand parliament speeches) and comparisons are made to our new human annotations. In the Economy topic, for example, out of 656 speeches total the model correctly identifies 389, corresponding to a recall of 0.59. The errors are somewhat evenly distributed, with the second-most-selected topic, Political System, having just 86 documents. In turn (looking to the Economy column), the most frequent topic that is mis-construed as Economy is also Political System. This likely reflects that these topics are often discussed in the same speech.

Overall, the results are quite encouraging about how well the model generalizes to the new corpus. Within each category, the correct class has by far the highest number of retrieved documents. The lowest recall of 0.426 (Freedom and Democracy) is almost identical to the lowest recall in the within-domain model (0.422). The recall does not get above 0.59, however. Because this data is at the speech level (rather than at the statement level as done in the source corpus), and speeches can consider multiple topics, this relative decrease in performance is perhaps not too surprising. The tendency for mis-classifications also looks different: while the within-domain model tends to put documents into Welfare, the cross-domain model tends to put documents into Political System.

Looking at the relative distribution of predictions (Panel B, Bottom Row), these also are quite encouraging. They are comparable to the within-domain model in the fidelity to the annotated distribution. But there are some interesting differences. While in the party platforms, Welfare is over-represented, in NZ parliament, Social Groups are over-represented. In addition, many more documents are identified as “No Topic” in the speeches, in disagreement with the human annotations.

In classification tasks with many labels, confusion matrices are not easy to read or fit on a page with legible text labels. One can observe this issue for our case in Appendix Section 3, which shows confusion matrices for the 44-topic specification. Here in the main text, we assess topic-level performance by reporting a number of topic-level metrics. These are reported in Table 4.

Table 4: Classifier Performance with 44 Topics

	Within-Domain				Cross-Domain				Ratio
	N	Top 1	Top 3	Top 5	N	Top 1	Top 3	Top 5	
Education	1810	0.769	0.908	0.948	177	0.746	0.910	0.955	0.970
Law and order	1387	0.696	0.864	0.925	158	0.715	0.892	0.943	1.027
Welfare state expansion	3644	0.794	0.943	0.975	368	0.685	0.897	0.948	0.863
Political authority	1021	0.470	0.724	0.830	775	0.570	0.831	0.895	1.213
Military	631	0.620	0.819	0.875	47	0.553	0.809	0.915	0.893
Environmental protection	1584	0.686	0.888	0.929	90	0.522	0.756	0.867	0.761
Underprivileged minority groups	388	0.222	0.495	0.624	10	0.500	0.800	0.900	2.256
Agriculture and farmers	775	0.573	0.787	0.862	87	0.494	0.713	0.816	0.863
Internationalism	667	0.537	0.792	0.865	37	0.486	0.676	0.784	0.906
Culture	596	0.584	0.787	0.837	43	0.465	0.698	0.791	0.797
Democracy	730	0.478	0.740	0.836	305	0.449	0.748	0.856	0.940
Economic growth	784	0.468	0.708	0.805	104	0.404	0.673	0.837	0.863
Technology and infrastructure	2183	0.710	0.901	0.948	113	0.398	0.628	0.788	0.561
Multiculturalism	440	0.448	0.698	0.780	103	0.398	0.631	0.709	0.889
Labour groups	950	0.576	0.804	0.860	188	0.383	0.681	0.766	0.665
Non-economic demographic groups	682	0.242	0.623	0.757	37	0.378	0.676	0.838	1.564
Nationalisation	168	0.429	0.607	0.714	32	0.344	0.531	0.656	0.802
Economic orthodoxy	478	0.479	0.667	0.768	136	0.331	0.566	0.728	0.691
Market regulation	883	0.411	0.676	0.787	114	0.298	0.553	0.719	0.725
Government / admin efficiency	1017	0.433	0.751	0.862	191	0.267	0.681	0.796	0.617
National way of life	666	0.341	0.620	0.748	61	0.262	0.689	0.787	0.770
Equality	1407	0.463	0.780	0.877	111	0.261	0.712	0.883	0.564
Protectionism	276	0.467	0.696	0.761	59	0.254	0.441	0.559	0.544
Centralization	734	0.407	0.688	0.790	52	0.250	0.654	0.712	0.614
Incentives	799	0.484	0.726	0.830	47	0.234	0.447	0.638	0.483
Traditional morality	439	0.390	0.677	0.777	67	0.194	0.388	0.522	0.498
Free market economy	419	0.239	0.525	0.654	73	0.096	0.233	0.397	0.402
Freedom and human rights	531	0.354	0.610	0.731	78	0.064	0.372	0.564	0.181
Political corruption	174	0.299	0.534	0.638	50	0.060	0.180	0.340	0.201
Civic mindedness	313	0.259	0.450	0.550	43	0.047	0.233	0.349	0.180
Constitutionalism	152	0.243	0.579	0.651	162	0.012	0.099	0.173	0.051
No topic	236	0.030	0.123	0.199	192	0.010	0.047	0.120	0.351
Anti-growth economy	576	0.255	0.674	0.776	13	0.000	0.308	0.462	0.000
Anti-imperialism	14	0.000	0.000	0.071	3	0.000	0.000	0.000	
Controlled economy	122	0.221	0.328	0.426	9	0.000	0.444	0.556	0.000
Corporatism/mixed economy	44	0.023	0.114	0.205	11	0.000	0.000	0.000	0.000
Economic goals	228	0.013	0.193	0.307	7	0.000	0.000	0.000	0.000
Economic planning	154	0.091	0.208	0.364	1	0.000	0.000	0.000	0.000
Foreign special relationships	182	0.264	0.637	0.714	5	0.000	0.000	0.000	0.000
Keynesian demand management	38	0.079	0.132	0.158	1	0.000	0.000	0.000	0.000
Middle class / professional groups	74	0.203	0.405	0.473	3	0.000	0.000	0.000	0.000
Peace	121	0.388	0.595	0.686	2	0.000	0.000	0.500	0.000
European Union	293	0.573	0.744	0.795					
Marxist analysis	43	0.070	0.209	0.326					
Total	28,853	0.538	0.763	0.839	4,165	0.410	0.649	0.746	0.762

In the table, each row is a topic, as indicated in the first column. Then there are two sets of columns corresponding to the within-domain and cross-domain classifier. Within these column groups, the first column (N) denotes the number of documents (statements or speeches, respectively) in the annotated test set. The remaining columns indicate topic-specific accuracy – Top 1, Top 3, and Top 5, respectively. As mentioned, Top 1 Accuracy is equivalent to class-level recall. Finally, the right-most column (Ratio) gives the ratio of cross-domain top 1 accuracy to within-domain top 1 accuracy. The table is sorted by cross-domain top 1 accuracy, from highest to lowest.

Overall, the 44-topic metrics produces a more mixed picture about the performance of our classifier. Even within-domain, there are some topics with quite poor performance. For example: Anti-Imperialism has zero accuracy, while Economic Goals has 0.01 accuracy. On the other hand, some topics are highly distinctive and easy to classify: Welfare State Expansion has 0.79 top 1 and 0.94 top 3 accuracy, for example, while the numbers for Education are 0.77 and 0.91. The other topics are somewhere in between, with the overall within-domain average accuracy being 0.538 (as already indicated in Table 2).

Some of these poor-performing categories can be explained by the Manifesto Project’s peculiar codebook choices. Topics like Anti-Imperialism (along with Corporatism / Mixed Economy, Keynesian Demand Management, Marxist Analysis, and Middle Class and Professional Groups) are rare (at least in English-language party platforms) and could probably be folded into other broader topics. Some topic pairs are difficult to distinguish semantically, for example Economic Goals vs Economic Growth, and the machine classifier tends to fold the smaller category into the larger one. It is unlikely that this type of subtle distinction would play an important role in downstream empirical applications.

For cross-domain, performance is slightly worse overall. Unsurprisingly, any topic that the within-domain classifier could not classify also performs poorly in the cross-domain classifier. Some topics (European Union and Marxist Analysis) do not show up in the NZ parliament speeches, so we cannot even compute metrics for them. Ten topics, which while infrequent, have zero accuracy. A number of other topics have quite poor performance, with the classifier even failing to rank the correct topic within the Top 5 a majority of the time. These metrics demonstrate the importance of some target-corpus validation, as machine-coded data on these poor-performing topics should not be used for any empirical analysis.

Looking at the top rows of the table (based on the sort), the cross-domain classification is quite good. The performance for Education (Top 1 Accuracy = 0.75, Top 3 = 0.91) is about the same as for the within-domain classifier. A handful of topics perform even better cross-domain than within-domain: Law and Order (Top 1 = 0.715), Political Authority (Top 1 =

0.685), Underprivileged Minority Groups (Top 1 = 0.5), and Non-Economic Demographic Groups (Top 1 = 0.38). Of the 44 topics, seven are ranked first correctly at least half the time, and all have good Top 3 / Top 5 Accuracy. For 24 topics, the correct topic is ranked within the Top 3 at least half the time.

The relevance of the variation will depend on the downstream empirical task. Fortunately, most of the categories with bad accuracy are quite rare, and thus they are unlikely to be the focus of empirical work. In our application below, we focus on Political Authority, which as mentioned is one of the better-performing topics for cross-domain learning.

4.3 Interpreting the Model Predictions

To further validate the cross-domain classifier, we examine interpretable dimensions of the predictions to check that it is working as expected. In particular, we desire that the classifier is using policy or issue language. We would not, for example, want it to be predicting off of the names of individuals or organizations whose mentions are spuriously correlated with topics.

First, we read the ten parliamentary speeches with the highest probability of belonging to each topic, using both the 44-topic and 8-topic specification. Appendix Section 6 includes text snippets from each of these speeches. In general, the speeches correspond very well to the specified topics and we do not see evidence that they are driven by correlated features.

Second, to systematically analyze the connection between text features and predicted topics, we produce a feature importance measure to identify the phrases that are positively correlated to topics in the target corpus. Appendix Section 7 describes in detail how this is done. Figure 1 illustrates word clouds of phrases that are positively related to the eight manifesto topics in the target corpus. As with the text snippets, the word clouds identify phrases that are clearly related to the associated topics. These provide some reassurance that our cross-domain classifier is capturing similar semantic dimensions in the target corpus.

4.4 Inter-Coder Reliability and Application to other Countries

Appendix Section 8 reports the results of our inter-coder reliability analysis. We find that the accuracy of our machine predictions is robust to different human coders.

The classifier also works quite well in a (smaller) corpus of U.S. congressional speeches. Using the classifier trained on 44 topics, we find that the top 1 accuracy is 0.44, and increases to 0.63 and 0.69 if we look at the top-3 and top-5 accuracies. In case we predict 8-topics, the top 1 accuracy increases to 0.52. These numbers are comparable to those computed using

Figure 1: Wordclouds of manifesto topics



the new hand annotations for New Zealand. These statistics are encouraging that the cross-domain classifier could work in other contexts besides the main application implemented here.

5 Empirical Application: Effect of New Zealand Electoral Reform

Previous research on parliamentary democracies and the political development in New Zealand suggest that the 1996 electoral reform from a first-past-the-post to a mixed-member proportional electoral system fundamentally changed parliamentary practices. In contrast to the pre-reform period, parties had to form coalition and minority governments, which are associated with principal-agent problems and lower stability (Martin and Vanberg, 2005; Powell, 2000; King et al., 1990; Barker et al., 2003; Vowles et al., 2002). Furthermore, the Standing Orders of the parliament were revised to explicitly reference parties and give them an important role in the allocation of speaking time (McGee, 2005; Proksch and Slapin, 2014). At the same time, parliamentarians had little experience with proportional representation and had to adapt to the new system (Taagepera and Shugart, 1989). Multiple parliamentarians also split from their party to form new parties.

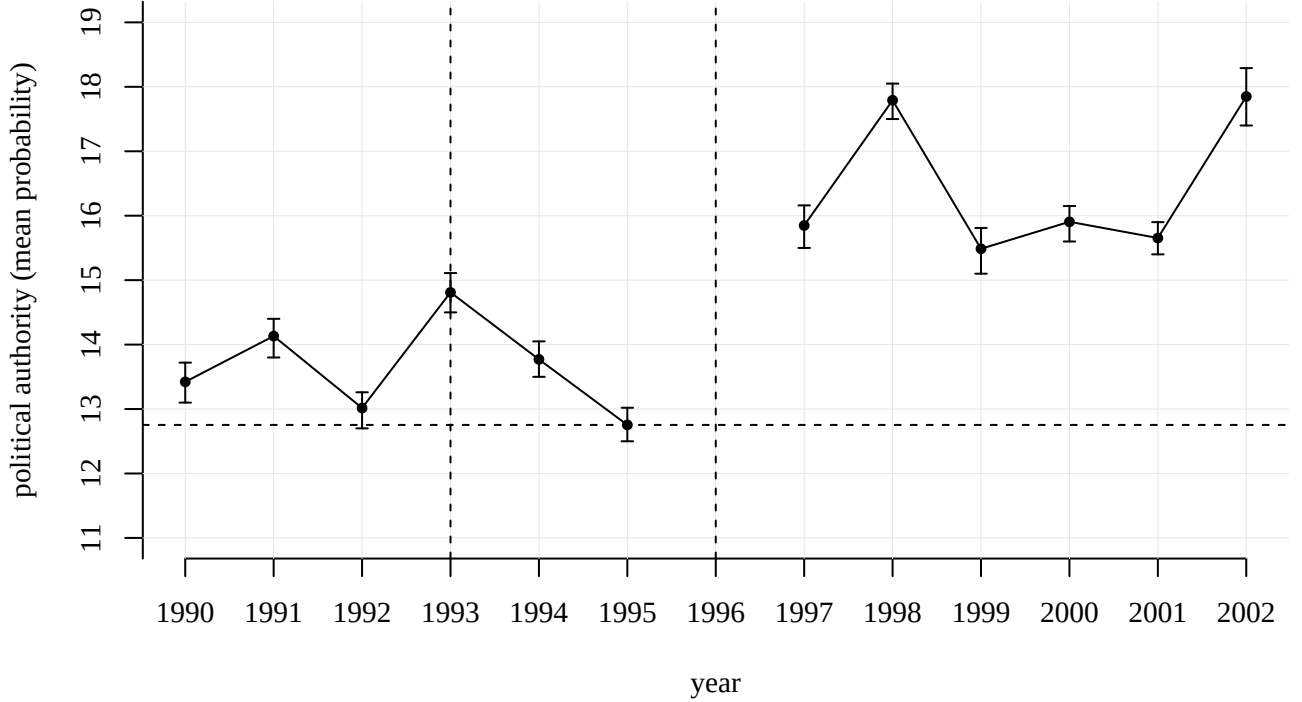
We assess the construct validity of our measures of topics by examining whether the allocation of attention reflects these fundamental changes in the New Zealand Parliament (Greene and Cross, 2017). We focus on the manifesto topic political authority, which covers issues related to stability and party competence. An example speech on this topic by Richard Prebble, leader of the ACT party, on 16 February 1999, is:

The Labour Party remains negative. I have seen seven Opposition leaders in my time but I have never seen a leader as relentlessly negative as Helen Clark. She must take lessons when she is in Africa. How could anybody be so negative, day in, day out? It could get into the Guinness Book of Records. She does not have a positive word to say about anything. It is all negative, negative, negative. The only plan Labour had was to hold a snap election so that it could get elected without any policy. The ACT party, by saying that we are opposed to instability and opposed to the “Italianisation” of New Zealand politics, has thwarted her plans.

We ask whether the reform increased attention toward political authority.

To show the effect of the reform on political authority, we show how it changed over time. Figure 2 plots the average probability that a speech focuses on political authority for

Figure 2: Effect of Electoral Reform on Political Authority



Note: Confidence intervals calculated using bootstrapped standard errors.

the years 1990 through 2002. We can see a clear discrete increase after the reform, relative to beforehand. The probability that a speech focuses on political authority increases about three percentage points from a pre-reform baseline of 13 percentage points.

Figure 2 appears to reflect well the political development in New Zealand. The average probability is highest in 1998 and 2002, which were two years marked by political instability and party politics. The first coalition government between the National party and New Zealand First broke down in 1998. In this context, multiple parliamentarians from New Zealand First left their party and continued to support the government. After the 1999 elections, Labour formed a government with Alliance, which terminated in 2002 when an early election took place. Also the founding leader of Alliance left his part in 2002 to form a new party, the Progressives (Barker et al., 2003; Vowles et al., 2002; Edwards, 2010).

We assess the statistical significance of the graphical evidence using a fixed-effects linear regression model. We estimate the effect of the reform using

$$y_{ijt} = \rho \text{PostReform}_t + \lambda_t + \gamma_j + \phi_j \cdot t + X'_{ijt} \beta + \varepsilon_{ijt} \quad (4)$$

Table 5: Regression Results: Effect of Reform on Political Authority

	(1)	(2)	(3)	(4)	(5)
Post Electoral Reform	0.268* (0.109)	0.243* (0.098)	0.297* (0.117)	0.247* (0.117)	0.194* (0.083)
Quadratic Trend	X	X	X	X	X
Speaker Fixed Effects		X	X		X
Speaker Trends			X		
Controls				X	
Weighting by Speech Length					X
N	290,456	290,456	290,456	290,456	290,456
R^2	0.008	0.092	0.110	0.138	0.093

Note: Standard errors, clustered by speaker, in parentheses. Appendix Table 10 includes all coefficients.+ $p < 0.10$, * $p < 0.05$, ** ($p < 0.01$)

where y_{ijt} is the log probability that speech i by speaker j at time t focuses on political authority. PostReform_t is an indicator variable equaling one for speeches held after the electoral reform, λ_t is a quadratic time trend, γ_j is a speaker fixed effect, $\phi_j \cdot t$ includes speaker-specific time trends, and $X'_{ijt}\beta$ includes other speech-level and speaker-level time-varying covariates. The covariates include indicator variables for gender, list MP, ministerial position, committee chair, and Maori constituency. They also include speech-level indicator variables for question, general debate, administrative speech, and committee stage, and a continuous variable for the timing in the governing cycle and party-level indicator variable for opposition party. Standard errors are clustered by speaker (although statistical tests are robust to two-way clustering by speaker and year).

The regression results are reported in Table 5. First, Model 1 includes the baseline model with just a constant and a quadratic time trend, Model 2 adds speaker fixed effects, and Model 3 includes speaker-time trends. We see a significant positive effect across specifications, consistent with the graphical results. According to the preferred specification Model 3, the probability that a specific speech corresponds to the topic political authority increases by about 30 percent after the reform. To provide some robustness, Model 4 includes in the regression the speech-level covariates. The effect of the reform is estimated to be smaller (19 percent) but still significant at the 0.05 level. Further, these results are robust to fully interacting these controls with the reform dummy. Finally, Model 5 is the same as Model 2 (speaker fixed effects), but the observations are weighted by length of the speech (in number of words). In all cases, the effect of the reform is robust and significant.

A number of supporting results are reported in the appendix. For one, we show similar results using a binary variable for whether political authority was the highest-share topic of a speech. Before the reform, political authority is the most important topic in 23 percent

of the speeches, and after the reform, that increases to 29 percent. The effect is driven by changes in the incumbent members, rather than selection of new types of members.

6 Concluding Remarks

This paper has studied cross-domain supervised learning as a new approach to classifying topics in political science. The method combines the low cost of unsupervised topic models with the high interpretability, scientific specificity, and validatability of supervised classifiers. In an era of large and growing public annotated datasets, we suspect that the applicability of this method will continue to expand.

We have demonstrated how to use this method in the context of the Comparative Manifesto Corpus and parliamentary speeches from New Zealand. We used a regularized multinomial regression model to learn a topic classifier in the source corpus and then apply it to predict topics in the target corpus. We showed how to validate the method using explanation methods in the target corpus, and more importantly using human annotations of a subset of target corpus documents.

To illustrate the empirical relevance of the method, we used our predicted topics to analyze the effects of the electoral reform in New Zealand, which changed in 1996 the electoral system from first-past-the-post to a mixed-member proportional representation system. In line with existing work on parliamentary democracies and New Zealand, we find that the electoral reform substantially increased attention toward political authority, which involves discussions about political instability and party competence.

Cross-domain supervised learning has the potential to increase our understanding of political phenomena such as responsiveness and accountability. An important advantage of the method is that one can estimate the same topics across documents and countries. For example, our approach allows us to study how closely the manifesto priorities match priorities in other documents such as speeches, party press releases, coalition agreements, legislative texts, and social media data. Moreover, cross-domain supervised learning can be used to improve the performance of existing measures of policy positions. For example, the tool could be used to identify non-ideological text or policy topics, which might improve the performance of existing methods (Slapin et al., 2018).

Future research may further improve the performance of cross-domain supervised learning by using alternative models or coding schemes and by providing additional training to coders. In our study, we did not provide any special training to coders and we used the exist-

ing manifesto coding scheme, which was not developed for cross-domain supervised learning. As political scientists have invested large resources in hand-coding data, we hope that our work encourages further research on transfer learning in political science research that uses text as data.

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Appendix:

Measuring Topics Using Cross-Domain Supervised Learning: Methods and Application to New Zealand Parliament

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1 Data

We use the Hansard as a source to identify the parliamentary speeches. This document offers a verbatim record of parliamentary speeches.¹ The New Zealand in-house service to report on debates was established in 1867 and in 1899 the reports became “substantially verbatim” (Edwards, 2015, p. 8). The goal of Hansard is to give the public un-biased information on parliamentary speeches. The name Hansard has its origins in England. Thomas Curzon Hansard compiled the debates of the House of Commons. In England, the Parliament took control of the reporting in 1909 (Ralphs, 2009, p. 8). The Hansard is an established source in political science and has mainly been used to study parliamentary speeches held in the House of Commons (Spirling, 2016). With few exceptions, scholars have not yet studied parliamentary debates in New Zealand using quantitative text analysis (e.g., Curran et al., 2018).

We access the data via the database provider *The Knowledge Basket*². The company provides us the speeches as html files. We wrote a python script to extract and segment the speeches. In addition, we identify relevant meta-data such as the stage of the speech, the date, the speaker name and type of speech. In the paper, we focus on the time period from 1987 until 2002. We remove speeches from the Speaker and Deputy Speaker and study speeches that include at least 40 characters.

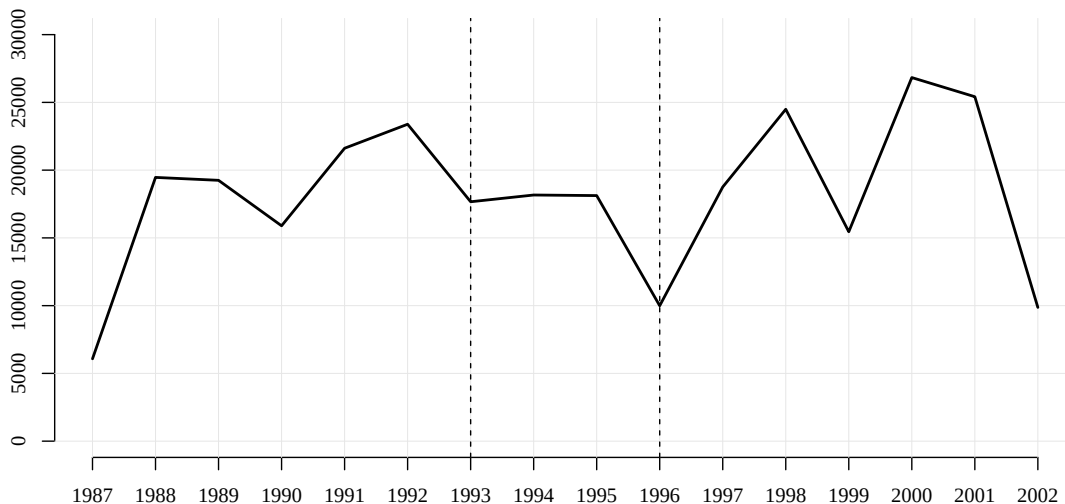
Our corpus takes into account 290,456 oral contributions including 154,438 speeches and 136,018 questions. Figure 1 illustrates the number of speeches by year. The number of speeches appears to be relatively constant over time with a slight increase after the 1996 electoral reform. We observe that in elections years the number of speeches is lower than in non-election years because the parliament has fewer sessions.

Figure 2 shows the number of speeches by party. We observe that the number of speeches correlates with the number of seats held in the New Zealand Parliament. The two parties with the largest number of speeches are the National and Labour parties followed by New Zealand First and ACT. Our data involves speeches from eight different parties: ACT, Alliance, Green, Labour, National, New Labour, NZ First, United New Zealand. The National party is a conservative and the ACT a liberal party. On the left side of the political spectrum, the New Zealand party system exhibits the Labour party as well as New Labour and the Alliance party. New Zealand

¹Additional information can be found on the webpage of the New Zealand Parliament: <https://www.parliament.nz/en/pb/hansard-debates/what-is-hansard/> (accessed on July 5, 2017).

²<http://www.knowledge-basket.co.nz> (accessed on July 7, 2017).

Figure 1: Number of speeches by year



Note: The dashed lines refers to the decision to introduce the reform (1993) and the first elections held under the new electoral system (1996).

First is a right-wing populist party and United New Zealand is located in the political center (Miller, 2005; Hayward, 2015, chapter 8). Note that we measure the party of parliamentary at the beginning of a legislative period.

We also collect data on individual-level characteristics of parliamentarians. We collect data from the Information Service of the New Zealand Parliament, the New Zealand Electoral Commission, and Woldendorp, Keman and Budge (2000). Our meta-data includes information on the gender, election mechanism, party membership, ministerial position, committee chairmanship, amongst others. Table 1 provides descriptive statistics on the variables used in the regression analysis.

Figure 2: Number of speeches by party

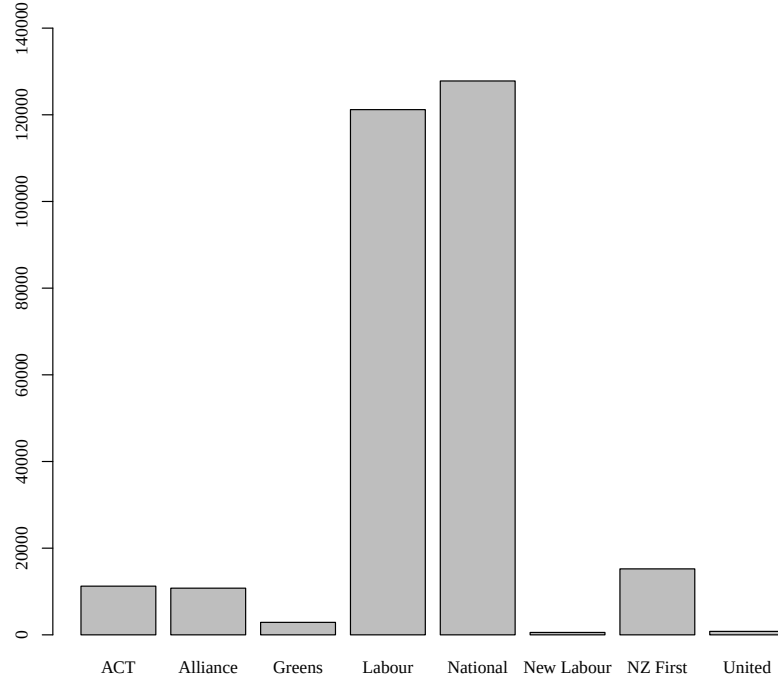


Table 1: Summary statistics

Variable	Mean	Std. Dev.	Min.	Max.
log_political_authority	1.681	1.704	-14.048	4.604
reform	0.417	0.493	0	1
list	0.181	0.385	0	1
female	0.173	0.378	0	1
ethnicity	0.082	0.275	0	1
minister	0.001	0.037	0	1
question	0.468	0.499	0	1
general_debate	0.044	0.206	0	1
admin	0.013	0.112	0	1
opposition	0.463	0.499	0	1
comchair	0.131	0.336	0	1
N	290456			

2 Manifesto Categories and Topics

We classify speeches to topics that are based on the Manifesto Project categories (Budge et al., 2001; Klingemann et al., 2006). In the following, we provide a list of manifesto categories. The detailed description of the manifesto categories can be found in the manifesto codebook.³

- per101 Foreign Special Relationships: Positive
- per102 Foreign Special Relationships: Negative
- per103 Anti-Imperialism
- per104 Military: Positive
- per105 Military: Negative
- per106 Peace
- per107 Internationalism: Positive
- per108 European Community/Union: Positive
- per109 Internationalism: Negative
- per110 European Community/Union: Negative
- per201 Freedom and Human Rights
- per202 Democracy
- per203 Constitutionalism: Positive
- per204 Constitutionalism: Negative
- per301 Decentralization
- per302 Centralisation
- per303 Governmental and Administrative Efficiency

³Manifesto Project Dataset. 2015. Codebook. <https://manifestoproject.wzb.eu/down/documentation> (accessed on November 10, 2017).

- per304 Political Corruption
- per305 Political Authority
- per401 Free Market Economy
- per402 Incentives: Positive
- per403 Market Regulation
- per404 Economic Planning
- per405 Corporatism/Mixed Economy
- per406 Protectionism: Positive
- per407 Protectionism: Negative
- per408 Economic Goals
- per409 Keynesian Demand Management
- per410 Economic Growth: Positive
- per411 Technology and Infrastructure: Positive
- per412 Controlled Economy
- per413 Nationalisation
- per414 Economic Orthodoxy
- per415 Marxist Analysis
- per416 Anti-Growth Economy: Positive
- per501 Environmental Protection
- per502 Culture: Positive
- per503 Equality: Positive
- per504 Welfare State Expansion
- per505 Welfare State Limitation

- per506 Education Expansion
- per507 Education Limitation
- per601 National Way of Life: Positive
- per602 National Way of Life: Negative
- per603 Traditional Morality: Positive
- per604 Traditional Morality: Negative
- per605 Law and Order: Positive
- per606 Civic Mindedness: Positive
- per607 Multiculturalism: Positive
- per608 Multiculturalism: Negative
- per701 Labour Groups: Positive
- per702 Labour Groups: Negative
- per703 Agriculture and Farmers: Positive
- per704 Middle Class and Professional Groups
- per705 Underprivileged Minority Groups
- per706 Non-economic Demographic Groups

For the 44-specification, we merged manifesto categories on the same topic, but different direction (positive/negative). For example, we combined the categories “per607 Multiculturalism: Positive” and “per608 Multiculturalism: Negative” to create one “Multiculturalism” topic.

For the 8-specification, we merged the categories to 8 larger topics following the manifesto codebook. These topics, denoted as domains in the codebook, are: external relations, freedom and democracy, political system, economy, welfare and quality of life, fabric of society, and social groups. We also add the no topic category, which the manifesto project codes as “0”. These are manifesto statements that could not be coded to the substantive categories.

3 Within-Domain Classification: Confusion Matrix

Table 3 presents the confusion matrix of the 44 topics. The training and test set come from the manifesto corpus. The training set included 75% of the English quasi-sentences of the manifesto corpus and the training set encompassed 25% of the corpus.

4 Within-Domain Classification: Precision and Recall

As described in the manuscript, we train our classifier based on the manifesto corpus. Our training data includes 75% of the English statements of manifesto corpus and the held-out test sample 25%. Hence, the training and test data are based on the manifesto corpus. In the following, tables we present the precision and recall by topic.

Table 2: Precision and recall (8 topics)

Topic	Precision	Recall
Economy	0.66	0.74
External relations	0.73	0.64
Fabric of society	0.62	0.57
Freedom and democracy	0.58	0.43
No topic	0.62	0.02
Political system	0.51	0.44
Social groups	0.59	0.42
Welfare and quality of life	0.67	0.77

Table 3: Precision and recall (44 topics)

Topic	Precision	Recall
Agriculture and farmers	0.61	0.57
Anti-growth economy	0.38	0.24
Anti-imperialism	1.00	0.04
Centralization	0.46	0.46
Civic mindedness	0.49	0.25
Constitutionalism	0.51	0.26
Controlled economy	0.61	0.22
Corporatism	0.33	0.02
Culture	0.70	0.57
Democracy	0.48	0.47
Economic goals	0.21	0.02
Economic growth	0.46	0.48
Economic orthodoxy	0.58	0.51
Economic planning	0.38	0.10
Education	0.69	0.77
Environmental protection	0.59	0.68
Equality	0.43	0.45
European Union	0.59	0.56
Foreign special relationships	0.38	0.19
Free market economy	0.39	0.23
Freedom and human rights	0.51	0.42
Governmental and administrative efficiency	0.44	0.46
Incentives	0.47	0.49
Internationalism	0.53	0.52
keynesian demand management	1.00	0.04
Labour groups	0.58	0.59
Law and order	0.60	0.71
Market regulation	0.49	0.44
Marxist analysis	0.83	0.10
Middle class and professional groups	0.63	0.30
Military	0.65	0.62
Multiculturalism	0.58	0.46
National way of life	0.37	0.35
Nationalisation	0.63	0.44
No topic	0.33	0.04
Non-economic demographic groups	0.47	0.25
Peace	0.54	0.41
Political authority	0.40	0.46
Political corruption	0.74	0.31
Protectionism	0.52	0.37
Technology and infrastructure	0.58	0.71
Traditional morality	0.54	0.39
Underprivileged minority groups	0.52	0.19
Welfare state expansion	0.55	0.78

5 Number of Speeches by Policy Areas and Legislative Period

Table 4 summarizes the number of speeches by topic and legislative period. More specifically, the table captures the most likely topic using the results from the 44-topic specification. Table 5 summarizes the absolute frequencies for the 8-topic specification.

Table 4: Number of speeches by topic and legislative period (44-topic specification)

topic	1987-1990	1990-1993	1993-1996	1996-1999	1999-2002
Agriculture and farmers	1787	1229	776	1199	914
Anti-growth economy	77	178	67	57	154
Centralization	1132	987	720	640	882
Civic mindedness	105	129	104	179	216
Constitutionalism	72	83	80	100	130
Controlled economy	44	19	12	20	22
Corporatism	2	0	0	0	2
Culture	425	588	428	318	388
Democracy	7792	7760	5436	7619	7716
Economic goals	27	22	16	17	13
Economic growth	1246	1531	839	697	1058
Economic orthodoxy	1893	1241	826	1012	616
Economic planning	13	21	13	8	26
Education	2436	3474	2738	2332	2892
Environmental protection	1559	1789	1097	1247	1777
Equality	1633	1900	1363	1682	1772
European Union	104	72	82	54	58
Foreign special relationships	19	27	16	28	37
Free market economy	298	391	231	367	376
Freedom and human rights	459	507	350	541	624
Governmental and administrative efficiency	4548	4474	3033	3308	3402
Incentives	975	640	251	553	563
Internationalism	413	420	412	435	633
Labour groups	1647	1813	984	1154	1798
Law and order	2958	3158	2244	3305	3208
Market regulation	1620	1430	1003	1683	1369
Middle class and professional groups	0	6	2	5	3
Military	671	466	434	475	1106
Multiculturalism	726	784	668	1093	469
National way of life	798	807	803	681	938
Nationalisation	645	251	188	381	296
No topic	436	400	271	350	377
Non-economic demographic groups	574	788	671	648	337
Peace	11	12	15	28	27
Political authority	12439	15118	10735	17379	18012
Political corruption	101	55	42	54	46
Protectionism	630	363	308	398	508
Technology and infrastructure	1799	1896	1239	1041	1888
Traditional morality	420	418	349	653	706
Underprivileged minority groups	256	267	224	263	266
Welfare state expansion	6037	8855	7142	6864	6555

Table 5: Number of speeches by topic and legislative period (8-topic specification)

topic	1987-1990	1990-1993	1993-1996	1996-1999	1999-2002
Economy	12377	10796	6749	8439	9601
External relations	1360	1104	1084	1128	1948
Fabric of society	5674	6474	5245	7600	7014
Freedom and democracy	6473	6576	4688	6888	7049
No topic	83	70	53	83	107
Political system	18104	20290	14380	20606	21386
Social groups	4261	4433	2862	3317	3389
Welfare and quality of life	10495	14626	11151	10807	11686

6 Text Snippets of Speeches by Topic

Table 6: Text snippets for 44 topics

Topic	Text
Agriculture and farmers	Some years ago there was a massive outbreak of eczema that depleted flocks and reduced farmers' incomes. That was followed by a drought of serious proportions that affected our flocks and herds of cattle. Then there was cyclone Bola—a disaster of unbelievable proportions. Even today, years after that event, farmers are still clearing up the mess, and, above all, are trying to pay the bills.
Anti-growth economy	Yes, it suggests two ways—progressive pricing and reduced fixed-line charges for the electricity supplied—but it basically does not rationally address the issue of getting greater levels of sustainable energy across the energy sector, nor does it make any suggestion to increase energy sustainability in the wider energy sense, such as relative to the percent of New Zealand's totally unsustainable energy requirement that is used by the transport sector.
Centralization	The Government insists that in local authorities with more than people ward systems will ensure that people in each local authority area will be represented. The Government will not have local authorities elected by just the more prosperous part of the community. It will have a system that will enable people to choose. Those people will not have to live in the area, but will need to be responsive to all of the communities in their area in order to function. Ward committees will be responsible for local interests in those particular areas.
Civic mindedness	The role of Business in the Community is to get the business sector involved in community projects. The project will not operate within the confines of a given community but will be a nationwide organisation. The organisation will be complementary to the efforts of community-based agencies such as the Hamilton enterprise agency. The organisation is supported by the community employment development unit. The person promoting the Business in the Community concept in New Zealand is Dr Graeme Craig of Hamilton, one of the founders of Woolrest International Ltd.
Constitutionalism	I rise to support the Bill. The issue is the shape and content of the constitution of this country, and I believe that nothing has changed since I studied constitutional law. As a young law student I was taught that the constitution resides with the people.
Controlled economy	What has the Government already done about the minimum wage?

Corporatism	The Government has received a welcome approach from Business New Zealand and the Council of Trade Unions to forge a tripartite social partnership to improve the quality and relevance of, and accessibility to, the workplace learning that is on offer. The Government respects the independence of both of our partners. We acknowledge that on some issues there are policy differences, but on the broad policy of workplace learning there is common ground. I welcome the approach by Business New Zealand and the Council of Trade Unions to work with the Government on those issues.
Culture	I move, That the Arts Council of New Zealand Bill be introduced. New Zealand's well-being is closely linked to the vitality of its cultural life. That is the reason that this Government supports the arts, as do all Governments in developed countries. It is now years since the Queen Elizabeth the Second Arts Council of New Zealand Act was passed.
Democracy	In little more than hours the people of the greatest democracy in the Americas, the United States, will vote for their new President. The man who will be responsible for leading the free world in the fight against socialism, and who will be responsible for ensuring the security of this planet, will be elected by, probably, one-eighth of the American people. In the United States—as in some other countries but certainly not in New Zealand or Australia—registration is not compulsory.
Economic goals	They show that unemployment rates in those regions have actually declined faster than in regions where unemployment rates were already rather lower. This Government has done exceptionally well in terms of lowering unemployment rates, both in deprived regions and amongst target groups in the community.
Economic growth	The OECD is optimistic about New Zealand's growth prospects. Its assessment is: "Over the coming years there will be a gradual but sustained growth in economic activity driven by exports, buoyant terms of trade, strong productivity gains, and a recovery in profits and business confidence."
Economic orthodoxy	What is the Reserve Bank's latest forecast for the Government's financial balance out-turn for the - fiscal year, and has he been advised whether it is consistent with his target of a Budget surplus?
Economic planning	The Alliance policy would be, first, to have an economic development strategy for New Zealand, so that we would be both beefing-up our social and economic infrastructures in the short term and developing a long-term plan. We would be looking at new technology-based industries.
Education	I recently visited a school in the West Coast area. I read the Education Review Office report prior to that visit, and I was taken aback by the comments made by the evaluator of the school about a particular teacher who, the report stated, was putting the education of the children at risk. She had an infant class, and the teacher who had those children after her had worked very hard every year for years to get those children up to scratch, up to the average level. She was going to resign because she was sick and tired of having to pick up after another teacher.
Environmental protection	I move, That the Wildlife (Penalties) Bill be now read a second time. I am pleased to be able to rise this evening to speak on my member's Bill, the Wildlife (Penalties) Bill, because its purpose is to increase the penalties for various offences under the Wildlife Act. But, more important, this Bill is about protecting the biodiversity of New Zealand. The Bill seeks to update the penalties for offences under the original Act, because the penalties have not been reviewed since . The Wildlife Act provides a protection system for wildlife, with some species absolutely protected and some partially protected.
Equality	How will those women receive pay justice after the abolition of the Employment Equity Act?

European Union	Is the Minister aware of the increasing centralisation of European trade issues in European Community bodies, such as the commission that the council administers and the European Parliament, and is it not Government policy to give priority to that area of our trade interests?
Foreign special relationships	I am delighted to bring the member up to speed, because the Government has been doing a lot of work in North Asia and on its relationship with North America. Sir Frank Holmes has been involved in that work.
Free market economy	It means that, if one holds on to a property and the property has a high capital cost—and one has allowed the person who is buying the property either to buy it or he or she is given that property free for the next years—there is a bonus in discounting in order to get the capital money in.
Freedom and human rights	I move, That the Human Rights Amendment Bill be now read a first time. The bill provides for a greater public sector accountability, in compliance with human rights obligations, and strengthens our human rights institutional framework and dispute-resolution procedures. This legislation is the long-awaited outcome of the Consistency audit project that was cancelled under the previous Government but revived under this Government.
Governmental and administrative efficiency	move, That the Department of Justice (Restructuring) Bill be introduced. The Bill continues the ongoing process of restructuring the Department of Justice. That restructuring has six main elements. A new Office of Treaty Settlements will replace the old Treaty of Waitangi policy unit, which was set up within the department on January . The office will be responsible directly to the Minister in charge of Treaty of Waitangi Negotiations, but will organisationally be part of the new Ministry of Justice.
Incentives	Mineral mining companies receive taxation advantages by way of concessions and incentives. Current incentives provide for a deduction of exploration and developmental expenditure. A company may also qualify for a deduction in advance for the amount of exploration and development expenditure it expects to incur in the next two income years. The tax payable by mining companies in respect of their income is two-thirds of the amount payable by other companies.
Internationalism	Kia ora, talofa lava and warm Pacific Island greetings. The thoughts and prayers of all New Zealanders will be with the people of the United States as they start to come to terms with the full extent of the tragedy that has unfolded today. This is a time for all of us to reflect on the vulnerability of nations, security, and life. My prayers go out to all who are suffering at this time. Speakers before me have touched on the many issues that this act of terrorism has brought forward.
Keynesian demand management	Will the Minister acknowledge that the increase in the goods and services tax was brought about by his inability to control spending, and that he cannot therefore visit his spending demands on others?
Labour groups	The real purpose of the Bill is to attack trade unions, and, specifically, to attack the New Zealand workers' union over the conduct of the union membership ballot. That is the excuse for the introduction of the Bill. The ballot was conducted under very difficult conditions, and I shall comment about that matter soon.
Law and order	I welcome this Bill and I am pleased that it has reached its second reading. It is appropriate to congratulate the Minister of Justice on the work that he has put into the Bill, with the support of the Minister of Police and members of the Government caucus, who have been right behind the Bill the whole way. The Bill tilts the balance of the justice system against the serious or persistent offender, and the rights of the public will now prevail over the rights of such offenders.

Market regulation	As I said, the Green Party remains very sceptical about industry self-regulation, given that no other country relies on it for the strategically important electricity industry. But we are supporting this bill, because we believe that it represents an improvement on the mess that the previous Government—Max Bradford, in particular—left us with. Self-regulation will lead to the major industry players wheeling and dealing around market rules to maximise their corporate benefit, and there are checks and balances to make sure that generators, retailers, and network companies hold each other to account in that process.
Middle class and professional groups	What other factors are impacting on low and middle income householders?
Military	I move, That this House take note of the Government's defence announcement. The Government today is announcing its defence plan to ensure that New Zealand has a modern, sustainable defence force matched to New Zealand's needs. Providing appropriately for the security and defence of the nation is a core responsibility of the Government.
Multiculturalism	The Government's current policy on providing Maori language and culture tuition in schools is clearly stated in Te Urupare Rangapu: "To provide for the Maori Language and culture to receive an equitable allocation of resources and a fair opportunity to develop having due regard to the contribution being made by Maori Language and culture toward the development of a unique New Zealand Identity."
National way of life	I am one of the old-fashioned Tory members of Parliament in this House who believes in God. I support the constitutional monarchy. In my lifetime, I do not want to see the flag changed, or the national anthem altered or shortened. I do not want to see the name of the country changed. I think that this country has seen so much change in recent times. We have changed the electoral system, for the worse. We are abandoning the Privy Council, for the worse. The royal honours system is under siege, for the worse. The republican debate in this country is for the worse.
Nationalisation	Will the sale of Trans Power assets "amount to a privatisation because many of the supply companies were now privatised" as stated by the Alliance energy spokesperson; if not, why not?
No topic	The Leader of the House has not discussed the details of this matter with me.
Non-economic demographic groups	What specific measures does the Minister intend to take to assist Maori women, given that the rate of unemployment for pakeha women is about one-quarter of the rate for Maori women?
Peace	There has been mixed progress in the peace process since the Townsville peace agreement last year. The international peace monitoring team, to which New Zealand contributes personnel—and Mr Sowry will remember that that issue came before the House to be discussed—has worked effectively. It has helped to sustain confidence in the peace process.
Political authority	It did not simply want a change in the person to whom the salary of the Prime Minister was paid; it wanted a real change away from sleaze, away from incompetence, and away from the appalling track record of this coalition Government. The country has got none of those things. The first thing that Mrs Shipley did was to wait a month before she assumed the prime ministership; a month of further inaction, a month of a lack of direction. Then, instead of doing what the country wanted her to do, which was to say to her junior coalition partner that it was the junior partner and that its antics would not be tolerated any longer in this country, Jenny Shipley continued to do what she had accused her predecessor of doing—swallowing dead rats. The only difference was that she did not wash them down with whisky.

Political corruption	No, they are not. The bribery and corruption referred to in the Crimes Act is the bribery and corruption as done by State servants. State servants are barred by this Act from accepting bribes in relation to their work or from corruption in relation to the Official Information Act.
Protectionism	They say that they are in favour of balance in their protection policy, but not yet. The Opposition wants to go to heaven, but is not prepared to die. This agreement is far from dying. The agreement has exceeded the expectations the Government had when it went into the negotiations. By July we will have established the world's most comprehensive free-trade area for goods, and the agreement that the Government reached on services goes beyond any other bilateral agreement in the world. I include the bilateral agreement negotiated last year between the United States and Canada.
Technology and infrastructure	The first stages of a number of key projects are progressing well. Developments such as the Grafton Gully connection to the port, the North Shore busway, and the "Spaghetti Junction" improvements should be ready for construction to start over the next months.
Traditional morality	They can't even agree on moral issues: homosexuality and abortion.
Underprivileged minority groups	What has he done to ensure that Pacific people's immigration concerns are being addressed?
Welfare state expansion	The debate today is the Appropriation (/ Financial Review) Bill, although one would be hard-pushed to tell that that is what it is about. Today, I want to talk about the key commitments that we have made in health. We came to office with seven key pledges that we made to the people of New Zealand, and one of those pledges was to focus on patients, not profits, and to cut waiting time for surgery.

Table 7: Text snippets for 8 topics

Topic	Text
Economy	In her statement on Tuesday the Prime Minister talked of the need for economic transformation. In discussing that, she talked further of the need for exports, modernisation, and expanding tourism. They are all very worthy ideals that we would support. There was not one word in that statement—and I have not heard one word from the Government since—about the transport systems that will be required to deliver those goals. Will these exports that will help to transform our economy all be moved around by e-commerce?
External relations	The Minister is very fiery, and he knows that morale among the troops is not high. I accept that the present Minister of Defence—perhaps unlike the former Minister of Defence—is interested in the troops.
Fabric of society	I move, That the Criminal Investigations (Blood Samples - Burglary Suspects) Amendment Bill be now read a first time. This bill will add a new weapon to the arsenal of the New Zealand Police to gather evidence against burglars, secure convictions, and expand DNA testing to burglary suspects. This bill will change the law to give the police the power to take compulsory DNA samples from burglary suspects in order to secure convictions.
Freedom and democracy	I welcome the debate on electoral reform that has been going on throughout New Zealand for the past year or so. I think that it is very good for the democratic process that people learn a lot more about their Parliament and how its members are elected. Indeed, the debate has enabled them to do that, although until now most of it has related to support for a change in the way in which members are elected, rather than an explanation of the present first-past-the-post system.
No topic	The Leader of the House has not discussed the details of this matter with me.

Political system	I shall just refer to the article to refresh my memory. That is permitted under the Standing Orders. The Contractors Federation (Inc.) said that it complained in a letter to the honourable member for Kaimai, who is the chairman of the select committee that the federation appeared before. That letter stated that the federation knew that its briefing to the select committee would be a circus and a farce and would have no effect on the decisions the Government intended to make.
Social groups	Yes, of all the farm workers throughout the country participated in the ballot. That is democracy Labour Government style. Even the Minister of Agriculture would acknowledge that if only workers out of participate in a ballot it is logical to assume that there is obviously no dissatisfaction on the part of workers with their conditions of employment.
Welfare and quality of life	Yes, it has and I shall talk a little about it. Teacher salaries make up about percent of total expenditure on education, so there is little to save unless one starts to hit teacher salaries. The Minister of Labour, the Minister of Education, and the Associate Minister of Education have all said that they want to use the Employment Contracts Act to slash teachers' wages and to reduce spending on education.

7 Wordclouds for Topics

We examine the phrases that are positively correlated to topics in the data of speeches. We start by extracting informative phrases from the speeches (see e.g., [Handler et al., 2016](#); [Denny and Spirling, 2018](#)). This approach allows us to recognize key phrases such as “new zealand”, and treat them as single tokens. In addition, we implemented the following text-preprocessing techniques to reduce the computational resources needed to calculate the word clouds. We tag parts of speech and identify phrases with up to four words using tag patterns, which results in a collection of noun and verb phrases.⁴ We remove upper-case and punctuation and then lemmatize the tokens to remove uninformative word endings. We filter out rare sequences that appear in fewer than 20 speeches or fewer than 30 times in total. We rank the phrases by their relative collocation (point-wise mutual information) to get key phrases.⁵ We filter out a set of policy-irrelevant words (e.g., names). The implemented vocabulary includes 20,956 words and phrases. Each speech is represented as a relative frequency distribution over that vocabulary.

⁴We use the following tag patterns: ‘A’, ‘N’, ‘V’, ‘P’, ‘C’, ‘D’, ‘AN’, ‘NN’, ‘VN’, ‘VV’, ‘NV’, ‘VP’, ‘NNN’, ‘AAN’, ‘ANN’, ‘NAN’, ‘NPN’, ‘VAN’, ‘VNN’, ‘AVN’, ‘VVN’, ‘VPN’, ‘ANV’, ‘NVV’, ‘VDN’, ‘VVV’, ‘NNV’, ‘VVP’, ‘VAV’, ‘VVN’, ‘NCN’, ‘VCV’, ‘ACA’, ‘PAN’, ‘NCVN’, ‘ANNN’, ‘NNNN’, ‘NPNN’, ‘AANN’, ‘ANNN’, ‘ANPN’, ‘NNPN’, ‘NPAN’, ‘ACAN’, ‘NCNN’, ‘NNCN’, ‘ANCN’, ‘NCAN’, ‘PDAN’, ‘PNPN’, ‘VDNN’, ‘VDAN’, ‘VVDN’. A: adverbs and adjectives, C: conjunctions, D: pronouns, N: nouns, P: prepositions, V: verbs.

⁵This is done by calculating the geometric mean of the pointwise mutual information criterion, since this metric can be calculated using the absolute rather than the relative frequencies. We set the following minimum levels: bigrams: 0.004, trigrams: 0.003, quadgrams: 0.002.

Then we calculate t-statistics on the correlation between phrase frequency and topic probabilities and plot phrases with the highest t-statistic.⁶

Figures 4, 5, 6, 7, and 8 illustrate wordclouds with phrases that are positively related to the probability that a speech belongs to one of the 44 topics. The wordclouds summarizes the phrases with the largest t-statistic.

⁶As our data is large, we focus on every 100th speech and the first 10,000 phrases.

Figure 4: Wordclouds for Topics I



Figure 5: Wordclouds for Topics II

(a) culture



(b) democracy



(c) economic goals



(d) economic growth



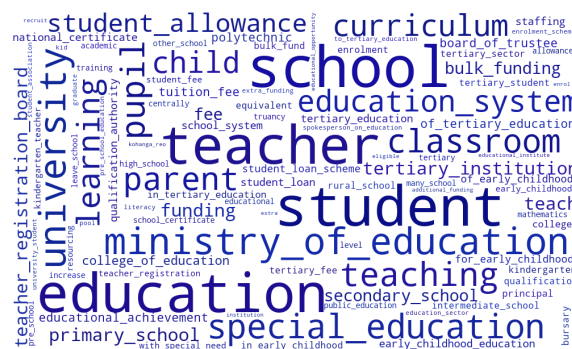
(e) economic orthodoxy



(f) economic planning



(g) education



(h) environmental protection



Figure 6: Wordclouds for Topics III

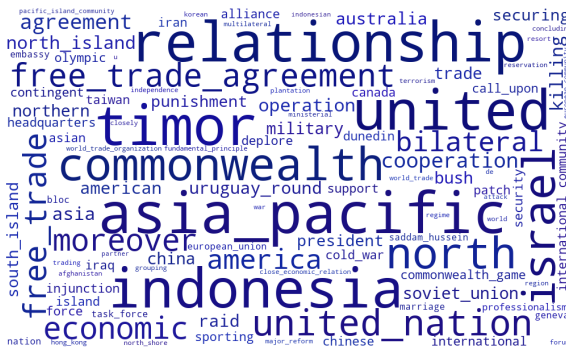
(a) equality



(b) european union



(c) foreign special relationships



(d) free market economy



(e) freedom and human rights



(f) governmental and administrative efficiency



(g) incentives

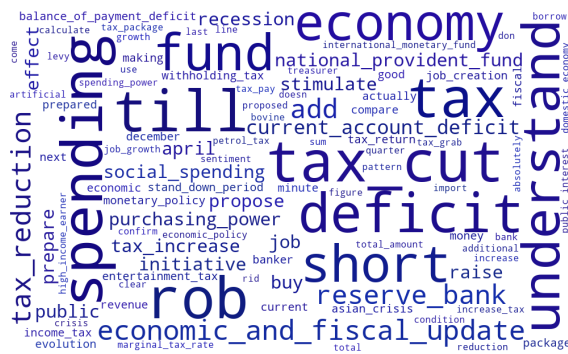


(h) internationalism



Figure 7: Wordclouds for Topics IV

(a) keynesian demand management



(b) labour groups



(c) law and order



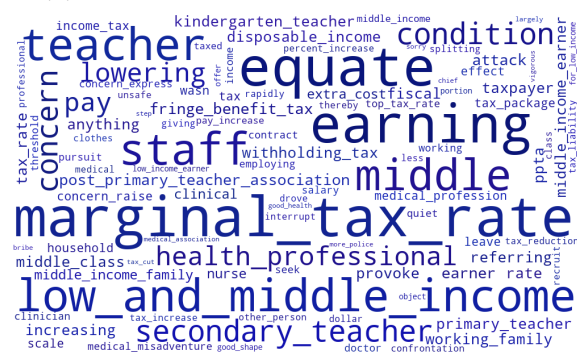
(d) market regulation



(e) marxist analysis



(f) middle class and professional groups



(g) military



(h) multiculturalism

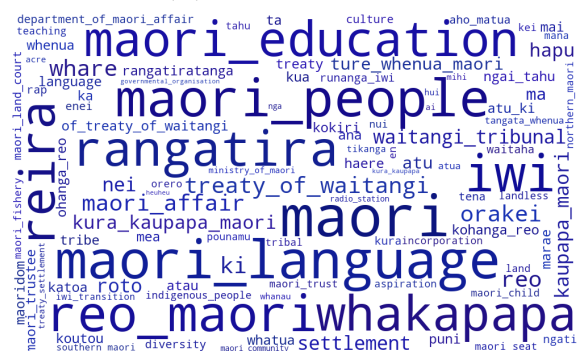


Figure 8: Wordclouds for Topics V

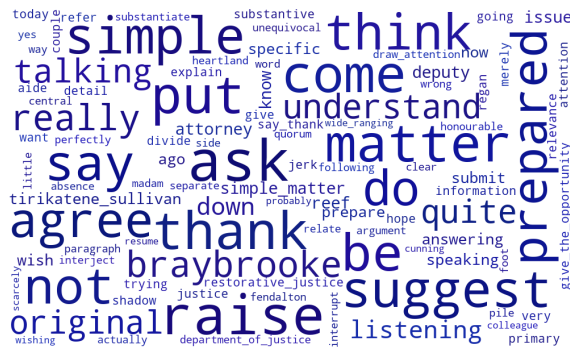
(a) national way of life



(b) nationalisation



(c) no topic



(d) non-economic demograpoc groups



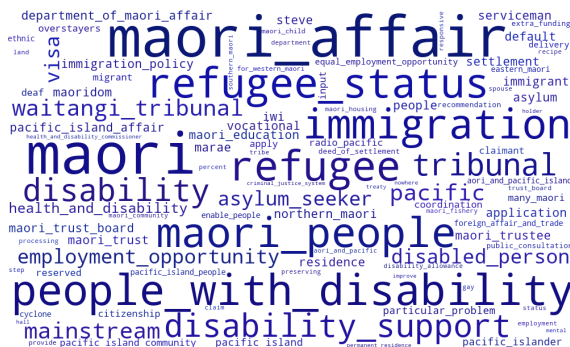
(e) peace



(f) political authority



(g) underprivileged minority groups



(h) welfare state expansion



8 Inter-coder Reliability

To understand how useful the target-corpus annotations are in assessing the validity of the classifier, we would like to have some sense of the error rate in the human codings. As [Mikhaylov, Laver and Benoit \(2012\)](#) show, the coder reliability of the manifesto project is relatively low.

To check this in our context, we hired three additional coders.⁷ Like the main coder, these coders also received training from the manifesto project in English-language platforms. They were not experts on New Zealand politics, however.

For this secondary annotation step, we drew a random sample of 250 speeches from the 4,165 speeches annotated by the first coder. Each of the three secondary coders annotated these same speeches, so that we had four annotations in total. We did not give the coders detailed guidelines, but asked them to code in line with their training from the manifesto project.

In Table 8, we compare the coding of these three secondary coders to the machine predictions and the annotations of the primary coder. The upper part of the table shows results for 44 topics, while the lower part shows results for 8 topics. In the first row, we find that in total the coders classified 37 percent of the speeches to the same category as the machine. If we focus on eight topics the accuracy is 49 percent, which is nearly identical to the accuracy that we obtained comparing the coding of the main coding to the machine predictions. In the next step, we compare the coding of the three coders to the coding of our main coder. In 50 percent of the speeches, the coders agreed on the same topic and this share increase to 63 percent if we limit ourselves to eight topics. Interestingly, these results are similar to [Mikhaylov, Laver and Benoit \(2012\)](#).

⁷We thank Pola Lehmann for providing us the contact details of the manifesto coders.

Table 8: Reliability of Coding

	Coder 1	Coder 2	Coder 3	Total
<i>44 Topics</i>				
Machine Predictions	0.364	0.388	0.368	0.373
Main Coder	0.452	0.528	0.508	0.496
<i>8 Topics</i>				
Machine Predictions	0.468	0.520	0.484	0.491
Main Coder	0.592	0.652	0.652	0.632

9 Details on the Electoral Reform

Table 9 summarizes the main differences between the electoral systems before and after the reform. In the first-past-the-post system, voters have one vote to select a candidate in their electorate. In the mixed-electoral proportional system, voters have two votes. Citizens can use the first vote to select a party at the national level and the second vote to select a candidate in the electorate. The reform increased the size of the parliament from 99 to 120 members. Notice that the size of the New Zealand varied in the period before the reform. The number of parliamentarians elected via district decreased from 95 to 60. The number of Maori districts increased from four to five with the option of further increases. Under the mixed-electoral system, the remaining (list) seats are allocated using the Saint Laguë formula. Similar to the electoral system in Germany, parties have to achieve at least 5 percent of the party votes or win an electorate (Barker et al., 2003; Vowles et al., 2002).⁸

Table 9: Comparison of First-Past-The-Post and Mixed-Member Electoral System.

	first-past-the-post	mixed-electoral system
number of votes	1	2
number of MPs	99	120 (+overhang)
number of districts	95	60
number of list MPs	95	55
number of Maori district	4	5
electoral rule (districts)	relative majority	relative majority
minimum entry criteria	win an electorate	5% party votes or win an electorate
formula for list seats	-	Saint Laguë formula

⁸See <http://www.elections.org.nz/> (accessed on July 30, 2017).

10 Economic Development in New Zealand

We want to make sure that the changes in political attention after the 1996 electoral reform were not caused by other events that took place at the same time. For this purpose, we checked whether any shock occurred that could have changed political attention after 1996.

First, we examine economic indicators from New Zealand. More precisely, a significant economic recession could have increased the division between National and Labour parliamentarians. Figure 9 illustrates the long-term development of four economic indicators: gross domestic product, gross domestic product (GDP) per capita, the unemployment rate and the inflation rate.⁹ The figures show the GDP, the GDP per capita and the inflation rate in the period 1982 until 2008. The unemployment rate is illustrated for the years between 1986 and 2008.

The gross domestic product has in 1996 around 70 billion US Dollars measured in current dollars. After the electoral reform, the GDP seems to slightly decrease, but then increases again. Hence, the average value of the gross domestic product in the legislative periods 1993-1996 and 1996-1999 is very similar. The trend in economic development seems to be in line with a normal long-term growth trend and not a period of economic crisis. The graph of the gross domestic product per capita, the consumer price index and the unemployment rate also do not provide evidence of a major economic crisis that could have substantially changed political attention after 1996. The inflation decreases after 1996 to about 1 percent, which is similar to the inflation rate in the early 1990s. The unemployment rate is in 1996 6.6 and then increases in 1999 to 7.7, but still remains at a low level that is similar to unemployment levels in the late 1980s.

Second, we controlled whether New Zealand underwent in 1996 any other substantial reform besides the electoral reform (Evans et al., 1996; Scott, 1996). Indeed, New Zealand government implemented important economic and public sector reforms since 1984. The reforms liberalized the markets (e.g., financial and labour markets) and aimed at making the public sector more efficient (e.g., privatization, reduction of employees). However, many of these reforms were already implemented in the 1980s or early 1990s. This timing suggests that the reforms cannot explain substantial changes

⁹Sources: The data on the GDP, GDP per capita and CPI stems from the Worldbank, and the data on the unemployment rate from NZ stats.

in political attention after 1996.

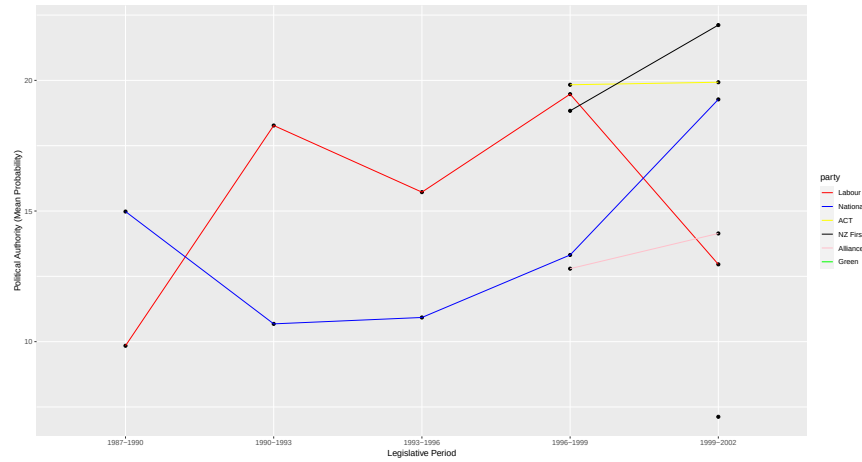
Figure 9: Economic indicators



11 Time Series Plot by Party

Figure 10 illustrate the average probability that a speech focuses on political authority by party and legislative period. We focus on party that parliamentarians belonged to at the beginning of a legislative term. The figure illustrates the that the average probability differ across periods and parties. For the National and Labour parties, the average probability appears to be related with their government status. In case the National or Labour party is in government, parliamentarians from the party talk less about political authority than when the party is in opposition. Opposition parties might have an especially strong incentive to talk about political authority to criticize the government. After the reform, both the main government and the main opposition parties talk more about political authority than thhe main government and opposition parties before the reform.

Figure 10: Change of political authority over time and across parties



12 Regression Models and Robustness

In the following, we present our results and robustness tests. The linear regression models use robust and clustered standard errors at the speaker level. Table 10 presents the main analysis with all covariates. Table 11 replicates the main results of our paper using a binary dependent variable and a logistic regression model.

Table 10: Regression analysis of political authority (main results)

	(1)	(2)	(3)	(4)	(5)
Post Electoral Reform	0.268* (0.109)	0.243* (0.098)	0.297* (0.117)	0.247* (0.117)	0.191* (0.084)
List MP					-0.095 (0.098)
Women					-0.377** (0.100)
Ethnic minority					0.247 (0.223)
Minister					0.063 (0.463)
Committee chair					-0.046 (0.074)
Question					-0.547** (0.034)
General debate					0.832** (0.047)
Administrative speec					0.070 (0.051)
Opposition party					0.502** (0.041)
Constant	1.486** (0.086)	1.419** (0.096)	1.158** (0.180)	1.277** (0.133)	1.545** (0.063)
<i>N</i>	290456	290456	290456	290456	290456
ll	-5.659e+05	-5.530e+05	-5.502e+05	-6.246e+05	-5.528e+05

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$ **Table 11:** Robustness II: Logistic regression analysis of political authority (binary dependent variable)

	(1)	(2)	(3)	(4)
Post Electoral Reform	0.286** (0.095)	0.241** (0.084)	0.256** (0.094)	0.223** (0.072)
Quadratic Trend	X	X	X	X
Speaker Fixed Effects		X	X	
Speaker Trends			X	
Controls				X
<i>N</i>	290,456	290,456	290,456	290,456
Pseudo R^2	0.053	0.045	0.052	0.052

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$

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